

Theory of phonon-magnon hybridization and angular momentum in CrI_3 and CrBr_3

Maxime Mignolet^{1,2,3*}, Miquel Royo^{4,5}, Massimiliano Stengel^{4,5} and Matthieu J. Verstraete^{1,3,6†}

¹ Nanomat Group, Q-Mat, Université de Liège, B-4000 Liège, Belgium

² Fonds de la Recherche Scientifique, B-1000 Bruxelles, Belgium

³ European Theoretical Spectroscopy Facility

⁴ Institut de Ciència de Materials de Barcelona (ICMAB-CSIC), Campus UAB, 08193 Bellaterra, Spain

⁵ ICREA-Institució Catalana de Recerca i Estudis Avançats, 08010 Barcelona, Spain

⁶ ITP, Department of Physics, University of Utrecht, 3508TA Utrecht, The Netherlands

★ M.Mignolet@uliege.be, † m.j.verstraete@uu.nl

Abstract

In magnetic materials angular momentum can be mediated by different carriers, including electrons, magnons, and phonons. The magnons can interact with circularly polarized phonons which are close in energy, provided specific symmetry conditions are met. If the interaction is strong enough the phonons and magnons shift in frequency and start to mix to form hybrid magneto-elastic quasi-particles. In this paper, we develop a constrained Hamiltonian framework which incorporates hybrid stiffness matrices and Berry curvatures. We quantify the degree of hybridization between phonons and magnons (up to 8% in CrI_3 and 25% in CrBr_3) using a decomposition of the total energy, which is a generalization of the norm decomposition for atomic contributions to phonons used in the literature. We also explore how the total angular momentum is conserved but shared between the phononic and magnonic subsystems upon hybridization.

Copyright attribution to authors.

This work is a submission to SciPost Physics.

License information to appear upon publication.

Publication information to appear upon publication.

Received Date

Accepted Date

Published Date

Contents

1	Introduction	2
2	Theory	3
2.1	Lagrangian and Hamiltonian mechanics	3
2.2	Second Quantization	6
2.3	Total energy and phonon-magnon hybridization	7
2.4	Nondimensionalization and reduction to pure phonons	9
2.5	Link to Raman spin-lattice interaction	10
3	Results	11
4	Conclusion	14

A Comparison of bare frequencies with experimental values	16
B Additional data	17
References	17

1 Introduction

Phonon angular momentum (PAM) has attracted broad interest since its prediction more than a decade ago [1]. It has come to light that phonon angular momentum plays a central role in many phenomena, such as the Einstein-de Haas effect [1], ultrafast demagnetization processes [2] and the thermal Hall effect [3–5]. Novel phenomena centered around PAM have also been unveiled, e.g. the phonon magnetic moment and the phonon Zeeman effect [6–9], the phonon diode effect [10] and the PAM Hall effect [11, 12]. Phonon angular momentum has been investigated as a mechanism for possible dark matter detectors [13], as well as for spin current injection in spintronics application [14]. PAM has been observed numerous times, both indirectly through phonon-assisted intervalley electron-hole recombination in WSe₂ [15], and directly through ultrafast electron diffraction in ferromagnetic Ni [2], by inelastic X-ray scattering in WC [16] and chiral quartz [17], and by measurement of the torque induced by generation of PAM by a temperature gradient in chiral Te [18].

Phonon angular momentum requires symmetry breaking to be well-defined and non-degenerate [19]. The breaking can involve the spatial inversion symmetry which can be found in non-centrosymmetric materials. It can be induced, for example, by a static external electric field [20–22], or driven by a time-dependent external electric field [23, 24]. The symmetry breaking can also concern time reversal symmetry, if the material itself is magnetic [19, 25], subject to an external magnetic field [6, 7], or to a rigid-body rotation [26].

Phonon angular momentum can interact and couple with other angular momenta, such as spins in magnetic materials. This interaction between PAM and spins can be modeled phenomenologically via a Raman spin-lattice coupling [27, 28]. Other modeling approaches use either an explicit magneto-elastic energy term, or a magnetic (Heisenberg) exchange coupling that depends on the interatomic distances [29–31].

The ab initio study of phonon angular momentum in magnetic materials goes through the inclusion of the molecular Berry curvature [19, 32], possibly including the dynamic of atomic magnetic moments [25, 33] into the equations of motion for lattice dynamics. The molecular Berry curvature is an adiabatic correction resulting from the contribution of the electronic cloud to the nuclear kinetic energy and canonical momentum [34, 35]. In the equations of motion, it manifests itself as a velocity-dependent force analogous to a (non-local) magnetic field.

In this paper, we develop a rigorous unified quantum theory of coupled phonon-magnon dynamics and hybridization. We treat the ionic and (atomic) spin degrees of freedom on the same footing. We reformulate the preexisting equations of motion to highlight particle-hole symmetry and the link with the total energy. We derive a novel orthonormalization scheme for the eigenmodes and establish a measure of the phonon-magnon hybridization, based on the partitioning of the total energy, which is one of the main results of this paper. Finally, we examine the exchange of angular momentum between magnons and phonons upon hybridization, which had not been quantified in the literature up to now.

The layout of the paper is as follows. Section 2 details the theoretical developments made,

which is subdivided in four parts: the derivation of the Lagrangian and Hamiltonian dynamics alongside the equations of motion, the second quantization of the framework, determining the total energy and quantifying the hybridization, and a discussion of the link between the current theory and Raman spin-lattice interaction. Section 3 presents applications of the theory to first-principles computations through the study of phonon-magnons in bulk CrI_3 and CrBr_3 . Lastly in section 4, we conclude and discuss possible future developments and applications.

2 Theory

2.1 Lagrangian and Hamiltonian mechanics

In this section, we will derive the equations of motion for coupled spin-nuclei dynamics. We denote $R = \{R_i\}$ the positions of the atoms. We define the mass-weighted displacement u_i of atom i from its equilibrium position R_{i0} as $u_i = \sqrt{M_i}(R_i - R_{i0})$. We note $s = \{s_i\}$ the deviations of the atomic spins from their equilibrium values S_{i0} , such that $s_i = S_i - S_{i0}$. We will consider a ferromagnetic material for which we have $S_{i0} = S_0 \forall i$. We use the notation $\dot{u}, \dot{s}$ to denote their time derivatives. $u, \dot{u}, s$ and $\dot{s}$ will constitute our dynamical variables. We consider that u and s evolve sufficiently slowly such that the electronic system remains in its ground state and we can write the wavefunction as a parametric function of u and s , which we denote $\Psi(r; u, s)$. The system's Lagrangian is given by

$$\mathcal{L}(u, \dot{u}, s, \dot{s}, t) = \frac{\dot{u}^2}{2} + \langle \Psi | i\hbar \frac{d}{dt} - H_e | \Psi \rangle \quad (1)$$

where H_e is the electronic Hamiltonian and includes the electron-electron, electron-ion and ion-ion Coulomb interactions, and the electronic kinetic energy. Considering that the dynamical variables are functions of time and using the chain rule, the total time derivative can be written as:

$$\frac{d}{dt} = \partial_t + \dot{u}_i \frac{\partial}{\partial u_i} + \dot{s}_i \frac{\partial}{\partial s_i}. \quad (2)$$

This yields the following Lagrangian in u and s :

$$\mathcal{L} = \frac{\dot{u}^2}{2} + i\hbar \dot{u}_i \langle \partial_{u_i} \rangle + i\hbar \dot{s}_i \langle \partial_{s_i} \rangle - V \quad (3)$$

where $V = \langle H_e \rangle$ and where we used the short-hand notation $\partial / \partial x = \partial_x$. We can perform a Legendre transform with respect to $u, \dot{u}$ and $s, \dot{s}$. The conjugate momenta are given by

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{u}_i} = \dot{u}_i + A_i^{(u)}, \quad (4)$$

$$\pi_i = \frac{\partial \mathcal{L}}{\partial \dot{s}_i} = A_i^{(s)}, \quad (5)$$

where we defined the atomic and spin Berry connections $A_i^{(u)} = i\hbar \langle \partial_{u_i} \rangle$, $A_i^{(s)} = i\hbar \langle \partial_{s_i} \rangle$ ¹. The first relation can be inverted to obtain an expression of the velocity $\dot{u}$ as a function of the position u and momentum p . However the same cannot be done for $\dot{s}$ with the second relation. Instead, we formulate the constraint $\phi_i = \pi_i - A_i^{(s)} \approx 0$. The " $\approx$ " sign means that the constraint should only be imposed after the equations of motion have been obtained [36–38]. The constrained Hamiltonian arising from the Legendre transform is written as:

$$\mathcal{H}(u, p, s, \pi, t) = p_i \dot{u}_i + \pi_i \dot{s}_i - \mathcal{L} + \lambda_i \phi_i \quad (6)$$

$$\Leftrightarrow \mathcal{H} = \frac{1}{2} (p_i - A_i^{(u)})^2 + V + \lambda_i \phi_i, \quad (7)$$

¹To lighten the equations, the factor $\hbar$ has been absorbed in the usual definition of the Berry connections.

where λ_i are Lagrange multipliers. The appeal of the Hamiltonian formulation over the Lagrangian is that it offers a natural path to obtain an expression for the total energy of the system and a decomposition of the eigenmodes in phononic, magnonic and coupled contributions. This will be shown in Sec. 2.3.

Since we started from an effective adiabatic Lagrangian formulation, the current approach does not include the Born contribution to the total energy [39, 40]. However the Born contribution does not break time reversal symmetry and is almost always neglected [25, 32, 33]. If needed, the Born contribution can be reincorporated while maintaining a Lagrangian starting point as detailed in Ref. [40].

From the above Hamiltonian we can derive the coupled equations of motion for the atomic spins and positions. Using Hamilton's equations, we obtain:

$$\dot{u}_i = \frac{\partial H}{\partial p_i} = p_i - A_i^{(u)}, \quad (8)$$

$$\dot{p}_i = -\frac{\partial H}{\partial u_i} = (p_j - A_j^{(u)}) \partial_{u_i} A_j^{(u)} - \partial_{u_i} V - \lambda_j \partial_{u_i} \phi_j, \quad (9)$$

$$\dot{s}_i = \frac{\partial H}{\partial \pi_i} = \lambda_j \partial_{\pi_i} \phi_j, \quad (10)$$

$$\dot{\pi}_i = -\frac{\partial H}{\partial s_i} = (p_j - A_j^{(u)}) \partial_{s_i} A_j^{(u)} - \partial_{s_i} V - \lambda_j \partial_{s_i} \phi_j. \quad (11)$$

We can now impose the constraint and use it to eliminate π . We can further use Eq. (10) to eliminate λ . We obtain:

$$\dot{u}_i = p_i - A_i^{(u)}, \quad (12)$$

$$\dot{p}_i = (p_j - A_j^{(u)}) \partial_{u_i} A_j^{(u)} - \partial_{u_i} V + \dot{s}_j \partial_{u_i} A_j^{(s)}, \quad (13)$$

$$\dot{A}_i^{(s)} = (p_j - A_j^{(u)}) \partial_{s_i} A_j^{(u)} - \partial_{s_i} V + \dot{s}_j \partial_{s_i} A_j^{(s)}. \quad (14)$$

If we place ourselves near an equilibrium reference configuration, the potential energy V and the Berry connections $A^{(u)}$ and $A^{(s)}$ can be expanded as Taylor series in u and s . The potential energy is expanded up to second order to retrieve the harmonic approximation. The Berry connections are expanded up to first order to obtain the Berry curvatures which are gauge-invariant:

$$V = \frac{1}{2} \begin{pmatrix} u & s \end{pmatrix} \begin{pmatrix} K^{uu} & K^{us} \\ K^{su} & K^{ss} \end{pmatrix} \begin{pmatrix} u \\ s \end{pmatrix}, \quad (15)$$

$$\begin{pmatrix} A^{(u)} \\ A^{(s)} \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} G^{uu} & G^{us} \\ G^{su} & G^{ss} \end{pmatrix} \begin{pmatrix} u \\ s \end{pmatrix}. \quad (16)$$

For simplicity, we assumed a symmetric gauge for the Berry connections [32]. The matrices G^{uu} , G^{ss} , and G^{us} and G^{su} are the molecular Berry curvature, the spin Berry curvature, and the mixed "spin-molecular" Berry curvatures respectively. Their general definition is given by

$$\begin{aligned} G^{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu \\ &= i\hbar \left(\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\nu \psi | \partial_\mu \psi \rangle \right) \end{aligned} \quad (17)$$

From this definition, it can be seen that the Berry curvatures G^{uu} and G^{ss} are real antisymmetric: $G^{uu} = -(G^{uu})^t$, $G^{ss} = -(G^{ss})^t$. G^{us} and G^{su} are real and the antisymmetric of one another:

$G^{us} = -(G^{su})^t$. The equations of motion become:

$$\dot{u}_i = \frac{1}{2}G_{ij}^{uu}u_j + p_i + \frac{1}{2}G_{ij}^{us}s_j, \quad (18a)$$

$$\dot{p}_i - \frac{1}{2}G_{ij}^{us}\dot{s}_j = -D_{ij}^{uu}u_j + \frac{1}{2}G_{ij}^{uu}p_j - D_{ij}^{us}s_j, \quad (18b)$$

$$\frac{1}{2}G_{ij}^{su}\dot{u}_j + G_{ij}^{ss}\dot{s}_j = D_{ij}^{su}u_j - \frac{1}{2}G_{ij}^{su}p_j + D_{ij}^{ss}s_j, \quad (18c)$$

where we introduced the following matrices:

$$D^{uu} = K^{uu} - \frac{1}{4}G^{uu}G^{uu}, \quad (19)$$

$$D^{us} = K^{us} - \frac{1}{4}G^{uu}G^{us}, \quad (20)$$

$$D^{ss} = K^{ss} - \frac{1}{4}G^{su}G^{us}. \quad (21)$$

And we have $(D^{uu})^t = D^{uu}$, $(D^{us})^t = D^{su}$, $(D^{ss})^t = D^{ss}$. The system of equations Eq. (18) can be recast in matrix form by multiplying the second line by minus one and permuting it with the first:

$$\mathcal{M}\dot{X} = \mathcal{D}X, \quad X = \begin{pmatrix} u \\ p \\ s \end{pmatrix} \quad (22)$$

with the matrices:

$$\mathcal{M} = \begin{pmatrix} 0 & -1 & \frac{1}{2}G^{us} \\ 1 & 0 & 0 \\ \frac{1}{2}G^{su} & 0 & G^{ss} \end{pmatrix}, \quad (23)$$

$$\mathcal{D} = \begin{pmatrix} D^{uu} & -\frac{1}{2}G^{uu} & D^{us} \\ \frac{1}{2}G^{uu} & 1 & \frac{1}{2}G^{us} \\ D^{su} & -\frac{1}{2}G^{su} & D^{ss} \end{pmatrix}. \quad (24)$$

Since the D^{xx} and G^{xx} matrices are real symmetric and antisymmetric, the matrices $\mathcal{M}$ and $\mathcal{D}$ are real antisymmetric and symmetric, respectively. Eq. (22) admits the general solution $X(t) = X e^{\Omega t}$. This yields the generalized eigenvalue problem

$$\Omega_\lambda \mathcal{M} X_\lambda = \mathcal{D} X_\lambda. \quad (25)$$

Since $\mathcal{D}$ is real symmetric and $\mathcal{M}$ is real antisymmetric, the eigenvalues are purely imaginary and come in complex conjugated pairs $i\omega$, $-i\omega$. There are $3 * N_{at} + N_s$ modes with positive frequency, and the same number with a negative frequency. The pairs bear the same physical meaning and correspond to the same physical solution. Furthermore, the corresponding eigenvectors X_λ also come in complex conjugated pairs. They can be $\mathcal{M}$ -orthonormalized through $X_\lambda^t \mathcal{M} X_{\lambda'} = \pm i \delta_{\lambda\lambda'}$. The choice for the prefactor will be made clear in Sec. 2.3.

From Eq. (22), (23) and (24), we can recover the equations for the bare phonons by discarding the spin parts and the molecular Berry curvature (which embodies adiabatic corrections):

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \dot{u} \\ \dot{p} \end{pmatrix} = \begin{pmatrix} D^{uu} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ p \end{pmatrix} \quad (26)$$

$$\Leftrightarrow \ddot{u} = -D^{uu}u. \quad (27)$$

This equation is the standard equation for phonons [41]. For spins, by discarding the ionic degrees of freedom, one obtains

$$G^{ss}\dot{s} = K^{ss}s. \quad (28)$$

This is the equation of motion derived by Niu and Kleinman [42], and has been shown to simplify to the Landau-Lifshitz equation under the assumption of rigid spins [43].

2.2 Second Quantization

We proceed to second quantize the equations of motion in Eq. (22). First, we perform a Fourier transform to make use of the periodicity of the problem:

$$\hat{x}_{l,\kappa} = \frac{1}{\sqrt{N}} \sum_q \hat{x}_{q,\kappa} e^{iqR_{l,\kappa}}, \quad x = \{u, p, s\} \quad (29)$$

The matrices transform accordingly. No mass factor appears here as u, p are already mass-scaled.

We now introduce the Holstein Primakoff transformation for the spins [44, 45]. For this, we switch from a cartesian representation to a $\pm$ ladder representation:

$$\begin{pmatrix} s_+ \\ s_- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \begin{pmatrix} s_x \\ s_y \end{pmatrix} = \mathcal{T}^{-1} \begin{pmatrix} s_x \\ s_y \end{pmatrix}, \quad (30)$$

where the q, κ indices have been omitted for clarity. This transformation is unitary. The matrices pertaining to the spins transform as:

$$\tilde{D}^{ss} = \mathcal{T}^{-1} D^{ss} \mathcal{T}, \quad \tilde{D}^{us} = D^{us} \mathcal{T}. \quad (31)$$

The Berry curvatures transform similarly. The pure phonon part of the hessian matrix (D^{uu}) and of the Berry curvature (G^{uu}) are left unchanged. This is a similarity transform of the $\mathcal{M}$ and $\mathcal{D}$ matrices. Therefore, it leaves the eigenvalues unchanged.

We can now introduce the Holstein-Primakoff transform:

$$s_{i,+} = \sqrt{S_0} \sqrt{1 - a_i^\dagger a_i / 2S_0} a_i, \quad (32a)$$

$$s_{i,-} = a_i^\dagger \sqrt{S_0} \sqrt{1 - a_i^\dagger a_i / 2S_0}, \quad (32b)$$

$$s_{i,z} = S_{i,z} - S_0 = -a_i^\dagger a_i, \quad (32c)$$

where $i = (q, \kappa)$. The Holstein-Primakoff operators obey the usual bosonic commutation relations: $[a_i, a_j^\dagger] = \delta_{ij}$, $[a_i, a_j] = [a_i^\dagger, a_j^\dagger] = 0$. Relations (32a), (32b) can be linearized within the large S limit [46, 47]:

$$s_{i,+} = \sqrt{S_0} a_i, \quad (33a)$$

$$s_{i,-} = \sqrt{S_0} a_i^\dagger. \quad (33b)$$

In order to diagonalize the Hamiltonian and solve the equations of motion, we use a Bogoliubov transform [37, 48–50]:

$$u_{q\kappa,\alpha} = \frac{1}{\sqrt{2}} (\epsilon_{q\kappa,\alpha} b_q + \epsilon_{-q\kappa,\alpha}^* b_{-q}^\dagger), \quad (34)$$

$$p_{q\kappa,\alpha} = \frac{1}{\sqrt{2}} (\mu_{q\kappa,\alpha} b_q + \mu_{-q\kappa,\alpha}^* b_{-q}^\dagger), \quad (35)$$

$$a_{q\kappa} = \frac{1}{\sqrt{2}} (\sigma_{q\kappa,+} b_q + \sigma_{-q\kappa,-}^* b_{-q}^\dagger), \quad (36)$$

where we made use of $u_{q\kappa,\alpha}^* = u_{-q\kappa,\alpha}$, $p_{q\kappa,\alpha}^* = p_{-q\kappa,\alpha}$, which stems from $u_{l\kappa,\alpha}, p_{l\kappa,\alpha}$ being real, to reduce the number of free parameters. The eigenvectors have components along u , p , and s which are ε , μ and σ . We note that $\varepsilon_{-q\nu,\kappa}^*$ does not reduce to $\varepsilon_{q\nu,\kappa}$ in general since time reversal symmetry is broken. The bosonic creation/annihilation operators obey the standard commutation relations: $[b_i, b_j^\dagger] = \delta_{ij}$, $[b_i, b_j] = [b_i^\dagger, b_j^\dagger] = 0$. The semiclassical equations of motion (22) now become:

$$-i\omega_{q\nu}\tilde{\mathcal{M}}\Xi_{q\nu}b_{q\nu} = \tilde{\mathcal{D}}\Xi_{q\nu}b_{q\nu}, \quad \Xi_{q\nu} = \begin{pmatrix} \varepsilon_{q\nu} \\ \mu_{q\nu} \\ \sigma_{q\nu} \end{pmatrix} \quad (37)$$

with the new matrices:

$$\tilde{\mathcal{M}} = \begin{pmatrix} 0 & -1 & \frac{1}{2}\tilde{G}^{us} \\ 1 & 0 & 0 \\ \frac{1}{2}\tilde{G}^{su} & 0 & \tilde{G}^{ss} \end{pmatrix}, \quad (38)$$

$$\tilde{\mathcal{D}} = \begin{pmatrix} D^{uu} & -\frac{1}{2}G^{uu} & \tilde{D}^{us} \\ \frac{1}{2}G^{uu} & 1 & \frac{1}{2}\tilde{G}^{us} \\ \tilde{D}^{su} & -\frac{1}{2}\tilde{G}^{su} & \tilde{D}^{ss} \end{pmatrix}, \quad (39)$$

and where we used $\dot{b}_{q\nu} = -i\omega_{q\nu}b_{q\nu}$. As it was the case for Eq. (25), the eigenfrequencies obtained from Eq. (37) come in pairs of positive and negative frequencies: $\omega_{q,-\nu} = -\omega_{q,\nu}$. It is customary to assign positive frequency solutions to the annihilation operator $b_{q\nu}$ and negative frequency solutions to the creation operator $b_{-q\nu}^\dagger$ [48]. In that sense, there is $3 * N_{at} + N_s$ independent modes, although there is double the amount of solutions to Eq. (37).

Before pursuing with the next sub-section, it is noteworthy to mention that the factors $\sqrt{S_0}$ introduced in the Holstein-Primakoff transformation Eq. (33) effectively brings a rescaling of the spin-related matrices as $\tilde{D}^{ss} \mapsto S_0\tilde{D}^{ss}$, $\tilde{D}^{su} \mapsto \sqrt{S_0}\tilde{D}^{su}$, and similarly for the other matrices (see Ref. [51]). In the case of non-equivalent spins $S_{i0} \neq S_{j0}$, the rescaling would become: $\tilde{D}_{ij}^{ss} \mapsto \sqrt{S_{i0}S_{j0}}\tilde{D}_{ij}^{ss}$, $\tilde{D}_{ij}^{su} \mapsto \sqrt{S_{i0}}\tilde{D}_{ij}^{su}$. This is in total analogy to the mass rescaling of the dynamical matrix in the standard theory of phonons [52, 53].

2.3 Total energy and phonon-magnon hybridization

We now turn our attention towards the total energy of the system. We restart from the Hamiltonian from Eq. (7). By using the Harmonic approximation for the potential energy and the Taylor expansion of the Berry connections from Eq. (15) and (16), we obtain the following expression for the total energy:

$$E = \frac{1}{2}X^t\mathcal{D}X. \quad (40)$$

The constraint does not contribute to the total energy due to it being identically zero for solutions satisfying the equations of motion. Plugging in the solutions from Eq. (25), we obtain the expression of the total energy in real space in the semi-classical regime:

$$E = \frac{1}{2} \sum_{\lambda} \Omega_{\lambda} X_{\lambda}^t \mathcal{M} X_{\lambda} \quad (41)$$

Using the second quantization procedure from the preceding sub-section, the total energy becomes

$$E = -\frac{1}{2} \sum_{q,\nu} i\omega_{q\nu} \Xi_{q\nu}^H \tilde{\mathcal{M}} \Xi_{q\nu} (b_{q,\nu}^\dagger b_{q,\nu} + 1/2), \quad (42)$$

where fluctuation terms of the form $b_{q\nu}b_{q\nu}$, $b_{q\nu}^\dagger b_{q\nu}^\dagger$ have been set to zero. This expression can be compared to the usual expression for the quantum harmonic oscillator: $E = \hbar\omega(b^\dagger b + \frac{1}{2})$. Identifying the former expression with the latter results in the normalization:

$$\Xi_{q\nu}^H \tilde{\mathcal{M}} \Xi_{q\nu} = 2i\hbar. \quad (43)$$

The total energy can be rewritten:

$$E = \sum_{q\nu} \hbar\omega_{q\nu} (b_{q\nu}^\dagger b_{q\nu} + 1/2). \quad (44)$$

With this, one can write the total energy for a single mode: $E_{q\nu} = i\omega_{q\nu} \Xi_{q\nu}^H \tilde{\mathcal{M}} \Xi_{q\nu} / 2 = \hbar\omega_{q\nu}$.

The $\tilde{\mathcal{M}}$ -normalization defined in Eq. (43), beyond establishing an orthonormalization of the eigenmodes, grants access to a natural decomposition of each mode into a phononic, magnonic, and coupled character based on their respective contributions to the total energy of the mode. The $\tilde{\mathcal{M}}$ -norm can be decomposed into

$$\begin{aligned} \Xi_{q\nu}^H \tilde{\mathcal{M}} \Xi_{q\nu} = & \left[(\varepsilon^H \quad \mu^H) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \varepsilon \\ \mu \end{pmatrix} \right. \\ & \left. + \sigma^H \tilde{G}^{ss} \sigma + (\varepsilon^H \tilde{G}^{us} \sigma + \sigma^H \tilde{G}^{su} \varepsilon) \right] \end{aligned} \quad (45)$$

From this decomposition, we can define the respective contributions to the total energy:

$$\eta_{q\nu}^{(\text{ph})} = \frac{1}{2i\hbar} (\varepsilon_{q\nu}^H \quad \mu_{q\nu}^H) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \varepsilon_{q\nu} \\ \mu_{q\nu} \end{pmatrix}, \quad (46a)$$

$$\eta_{q\nu}^{(\text{mag})} = \frac{1}{2i\hbar} \sigma_{q\nu}^H \tilde{G}^{ss} \sigma_{q\nu}, \quad (46b)$$

$$\eta_{q\nu}^{(\text{cpld})} = \frac{1}{2i\hbar} (\varepsilon_{q\nu}^H \tilde{G}^{us} \sigma_{q\nu} + \sigma_{q\nu}^H \tilde{G}^{su} \varepsilon_{q\nu}), \quad (46c)$$

with the different contributions summing to one

$$\eta_{q\nu}^{(\text{ph})} + \eta_{q\nu}^{(\text{mag})} + \eta_{q\nu}^{(\text{cpld})} = 1. \quad (47)$$

The energy for a single mode now becomes:

$$E_{q\nu} = \hbar\omega_{q\nu} [\eta_{q\nu}^{(\text{ph})} + \eta_{q\nu}^{(\text{mag})} + \eta_{q\nu}^{(\text{cpld})}] \quad (48)$$

This decomposition of hybrid modes into different characters (phononic, magnonic, coupled) is an improvement with respect to the current literature where no clear method surfaces, and the spin norm choice is often not explicit [54–57]. The phononic character can be further decomposed into atomic contributions

$$\eta_{q\nu}^{(\text{ph})} = \frac{1}{2i\hbar} [\mu_{q\nu,\kappa}^H \varepsilon_{q\nu,\kappa} - \varepsilon_{q\nu,\kappa}^H \mu_{q\nu,\kappa}], \quad (49a)$$

$$\eta_{q\nu}^{(\text{ph})} = \sum_{\kappa} \eta_{q\nu,\kappa}^{(\text{ph})}. \quad (49b)$$

Such an atomic-resolved decomposition is generally not possible for the magnonic and the coupled character. However, it may still provide a good physical insight. Within the approximation of rigid atomic spins, the spin Berry curvature $\tilde{G}^{ss}$ is block diagonal and the magnonic character can be decomposed exactly in atomic-resolved contributions [43].

Incidentally, $\eta_{q\nu}^{(\text{mag})}$ corresponds also to the reduction $\Delta S_z = S_{0,z} - \langle S_z \rangle$ of the spin z -component with respect to the ground state due to the excitation of the mode $q\nu$ [43]. In consequence, the spin angular momentum carried by a magnon is

$$l_{q\nu}^{(\text{mag})} = \hbar \eta_{q\nu}^{(\text{mag})} b_{q\nu}^\dagger b_{q\nu}. \quad (50)$$

Additionally to magnons, phonons can also carry angular momentum which in this case originates from the collective rotation of atoms around their equilibrium position. The canonical angular momentum of a single atom can be written classically $l_i = u_i \times p_i$. Introducing the second quantization procedure from Section 2.2, we obtain the expression for the phonon angular momentum

$$l_{q\nu}^{(\text{ph})} = \text{Re}\{\varepsilon_{q\nu,\kappa}^* \times \mu_{q\nu,\kappa}\} (b_{q\nu}^\dagger b_{q\nu} + \frac{1}{2}). \quad (51)$$

2.4 Nondimensionalization and reduction to pure phonons

We now turn to the special case where phonons do not interact with magnons, but still incorporate effects of the time reversal symmetry breaking through the molecular Berry curvature G^{uu} . In this setting, the equations of motion Eq. (37) become

$$-i\omega_{q\nu} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \varepsilon_{q\nu} \\ \mu_{q\nu} \end{pmatrix} = \begin{pmatrix} D^{uu} & -\frac{1}{2}G^{uu} \\ \frac{1}{2}G^{uu} & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_{q\nu} \\ \mu_{q\nu} \end{pmatrix}. \quad (52)$$

Up to now, the eigenvectors $\varepsilon_{q\nu}$, $\mu_{q\nu}$ associated to the atomic displacements and momenta carried dimensions the dimensions of a length times the square root of a mass $LM^{1/2}$, and the inverse $L^{-1}M^{-1/2}$. To nondimensionalize these quantities we can define a characteristic mass-scaled length $l_{q\nu}^0$ such that $\varepsilon_{q\nu} = l_{q\nu}^0 \tilde{\varepsilon}_{q\nu}$, $\mu_{q\nu} = \tilde{\mu}_{q\nu}/l_{q\nu}^0$. We will further impose $\|\tilde{\varepsilon}_{q\nu}\| = 1$. Multiplying the second line of Eq. (52) by $\varepsilon_{q\nu}^H$ on the left, one obtains

$$-i\omega_{q\nu} \varepsilon_{q\nu}^H \varepsilon_{q\nu} = \frac{1}{2} \varepsilon_{q\nu}^H G^{uu} \varepsilon_{q\nu} + \varepsilon_{q\nu}^H \mu_{q\nu}. \quad (53)$$

Subtracting the complex conjugate and rearranging the terms, one gets

$$\omega_{q\nu} \varepsilon_{q\nu}^H \varepsilon_{q\nu} - \frac{i}{2} \varepsilon_{q\nu}^H G^{uu} \varepsilon_{q\nu} = \frac{1}{2i} (\varepsilon_{q\nu}^H \mu_{q\nu} - \mu_{q\nu}^H \varepsilon_{q\nu}). \quad (54)$$

Using the normalization condition (43) and replacing $\varepsilon_{q\nu}$ by its nondimensional form $\tilde{\varepsilon}_{q\nu}$, we finally obtain the characteristic length

$$l_{q\nu}^0 = \sqrt{\frac{\hbar}{\omega_{q\nu} - \frac{i}{2} \varepsilon_{q\nu}^H G^{uu} \varepsilon_{q\nu}}}. \quad (55)$$

In the case on time reversal symmetric phonons, it reduces to $l_{q\nu}^0 = \sqrt{\frac{\hbar}{\omega_{q\nu}}}$. Furthermore the definition of the atomic displacements Eq. (34) becomes

$$u_{q\kappa,\alpha} = \sqrt{\frac{\hbar}{2\omega_{q\nu}}} (\tilde{\varepsilon}_{q\kappa,\alpha} b_q + \tilde{\varepsilon}_{-q\kappa,\alpha}^* b_{-q}^\dagger) \quad (56)$$

Both of these results agree with the textbook definition [52].

We now show that the decomposition of the mode energy as well as the atomic-resolved mode decomposition of the phononic character defined in Eq. (49) is a generalization to the

decomposition usually performed in the literature for pure time reversal symmetric phonons [58]. Since we are dealing with purely phononic modes, we have $\eta_{qv}^{(\text{ph})} = 1$. Restarting from Eq. (49a) and extracting μ_{qv} from the second line of Eq. (52), one obtains

$$\eta_{qv,\kappa}^{(\text{ph})} = \varepsilon_{qv,\kappa}^H (\omega_{qv} - iG^{uu}/2) \varepsilon_{qv,\kappa} / \hbar. \quad (57)$$

Inserting the nondimensionalized $\tilde{\varepsilon}_{qv}$ and using (55), the atomic-resolved mode decomposition reduces to

$$\eta_{qv,\kappa}^{(\text{ph})} = \|\tilde{\varepsilon}_{qv,\kappa}\|^2, \quad \eta_{qv}^{(\text{ph})} = \sum_{\kappa} \eta_{qv,\kappa}^{(\text{ph})} = 1. \quad (58)$$

This decomposition is valid for both time reversal symmetric and non time reversal symmetric phonons without phonon-magnon interaction. It matches the decomposition used in the literature for time reversal symmetric phonons [58].

2.5 Link to Raman spin-lattice interaction

In this paragraph we establish the link between the molecular Berry curvature and the Raman spin-lattice interaction, which was already discussed by [32, 59]. Since the molecular Berry curvature (MBC) is non-zero only in systems where the $\mathcal{T}$ symmetry is broken and that, in ferromagnetic materials, the symmetry breaking is due to its magnetization $\vec{M}$, it is natural to write the MBC as a function of $\vec{M}$: $G^{uu} = G^{uu}(\vec{M})$. Since the ground states with $+\vec{M}$ and $-\vec{M}$ magnetizations are linked by time reversal symmetry, we have $G^{uu}(\vec{M}) = -G^{uu}(-\vec{M})$. To first order in $\vec{M}$ the MBC can be expanded as

$$G_{ij}^{uu}(\vec{M}) = G_{ij\delta}^{uu} M_{\delta}. \quad (59)$$

Injecting this expression into the Hamiltonian of Eq. (7) and isolating the term proportional to p and u , one obtains:

$$H_{pu} = \frac{1}{2} \sum_{ijk} G_{ij\delta}^{uu} p_i u_j M_{\delta} \quad (60)$$

$$= \frac{1}{2} \sum_{ll'} \sum_{\kappa\alpha, \kappa'\beta, k} G_{l\kappa\alpha, l'\kappa'\beta, \delta}^{uu} p_{l\kappa\alpha} u_{l'\kappa'\beta} M_{\delta}. \quad (61)$$

This term is non-local in general. Restricting ourselves to the on-site contribution ($l\kappa = l'\kappa'$), we can make use of the skew-symmetry of the MBC to write $G_{l\kappa\alpha, l\kappa\beta, k}^{uu} = \varepsilon_{\alpha\beta\gamma} g_{l\kappa, \gamma, k}^{uu}$. This effectively transform the matrix-vector product as a vector cross-product. Developing the above expression gives:

$$H_{pu}^{\text{loc}} = \frac{1}{2} \sum_{l\kappa, \alpha\beta\gamma, \delta} p_{l\kappa\alpha} (\varepsilon_{\alpha\beta\gamma} g_{l\kappa, \gamma, \delta}^{uu}) u_{l\kappa\beta} M_{\delta} \quad (62)$$

$$= \frac{1}{2} \sum_{l\kappa, \delta} \mathbf{p}_{l\kappa} \cdot (\mathbf{g}_{l\kappa, \delta}^{uu} \times \mathbf{u}_{l\kappa}) M_{\delta} \quad (63)$$

$$= \frac{1}{2} \sum_{l\kappa, \delta} (\mathbf{u}_{l\kappa} \times \mathbf{p}_{l\kappa}) \cdot \mathbf{g}_{l\kappa, \delta}^{uu} M_{\delta} \quad (64)$$

$$= \frac{1}{2} \sum_{l\kappa, \gamma\delta} L_{l\kappa\gamma} g_{l\kappa, \gamma\delta}^{uu} M_{\delta}. \quad (65)$$

Further assuming an isotropic g^{uu} , this can be simplified into:

$$H_{pu}^{\text{loc}} = \frac{1}{2} \sum_{l\kappa} g_{l\kappa}^{uu} \mathbf{L}_{l\kappa} \cdot \mathbf{M}, \quad (66)$$

which is the expression of the Raman spin-lattice interaction under the mean field approximation [27].

3 Results

We performed *Density Functional Theory* calculations on the ferromagnetic semiconducting Van der Waals materials CrI_3 and CrBr_3 [60, 61]. We used the ABINIT software [62, 63] with the Perdew and Wang correlation functional [64]. We used a planewave basis set with an energy cutoff of 40 Ha and a $6 \times 6 \times 6$ k -point grid. The lattice parameter and atomic positions have been relaxed to achieve forces on atoms under 10^{-6} Ha/Bohr. We used constrained magnetic moment DFPT to perform calculations at fixed spin and obtain derivatives with respect to atomic spins [65]. All calculations were performed at $q = \Gamma$.

CrI_3 and CrBr_3 belong to the space group $R\bar{3}$ and possess the spatial inversion symmetry (however they do not have time reversal symmetry due to their magnetization). At the Γ point, the phonons modes can be decomposed into 4 modes with irreducible representations (irreps) A_g , 4 E_g , 4 pairs of doubly-degenerate A_u , and 4 pairs of doubly-degenerate E_u . One A_u mode and one pair of E_u modes are acoustic phonons, while the rest are optical phonons [25, 33]. Since CrI_3 and CrBr_3 have two magnetic Cr ions per unit cell, they have one acoustic (AM) and one optical (OM) magnon mode each.

The bare phonon and magnon frequencies can be obtained by solving the uncoupled equations of motion Eq. (27) and (28). In this setting, the angular momentum of phonons is zero and the angular momentum of magnons is equal to $\hbar$. The obtained bare phonon frequencies are in correct agreement with Raman scattering experiments and previous DFT calculations [66–69]. A comparison table can be found in Appendix A. The optical magnon frequencies predicted here at 28 and 19 meV respectively for CrI_3 and CrBr_3 deviate significantly from the experimental values of 19 and 13 meV [70, 71]. This is a known limitation of DFT when using an LDA functional [72] and is consistent with previous DFT calculations [65, 72, 73].

The coupled phonon-magnon frequencies are obtained by solving the full equations of motion (37). The bare and coupled frequencies are reported in Fig. 1(a) for CrI_3 and Fig. 2(a) for CrBr_3 . From the molecular Berry curvature and the coupling with the magnons, the phonons are able to obtain non-zero angular momentum, which is reported in Figs. 1(b), 2(b). Up to numerical accuracy, non-degenerate phonons (with A_g or A_u irreps) undergo no shift in frequency or hybridization with the magnons. Doubly-degenerate phonons (with E_g or E_u irreps) are split in frequency by the inclusion of the molecular Berry curvature and the coupling with magnons. Each pair is split into a left and a right circularly polarized mode. The largest frequency splitting occurs in the E_u phonon mode pair the closest in frequency to the optical magnon. This is in agreement with the conclusions of Ref. [33]. The values of the frequency splittings and the phonon angular momentum are summarized in table 1 and 2.

Using Eq. (46a), (46b) and (46c), we can compute the phononic, magnonic and hybrid character of each mode. In CrI_3 , all phonons remain pure to within $\pm 0.1\%$, except one of the mode in the E_u pair at 26.81 meV (see Table 3). The phonon mode hybridizes and obtains $\sim 8\%$ magnonic character, and the optical magnon incorporates around the same amount of phononic character. Similarly in CrBr_3 , all phonon modes remain phonon-like within $\pm 0.1\%$, except for one of the E_u modes at 19.73 meV, which gains $\sim 25\%$ magnonic character. This large fraction is exchanged with the optical magnon which itself recovers 25% phononic char-

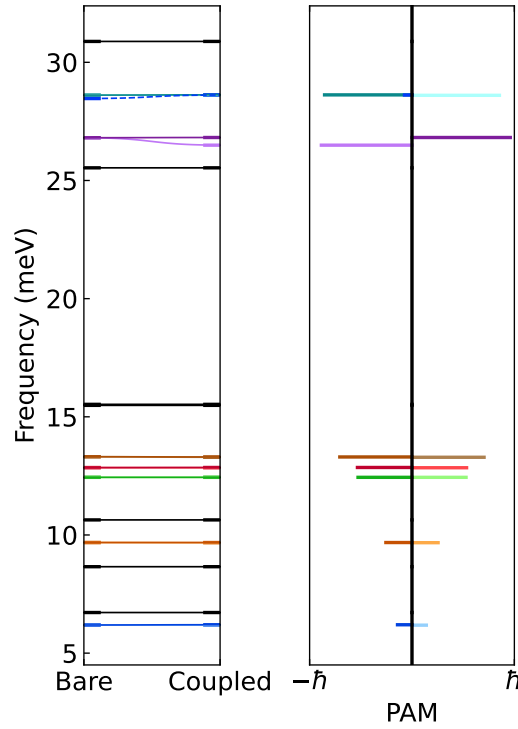


Figure 1: Frequency splitting and phonon angular momentum for CrI3 at the Γ point. (a) Bare and coupled frequencies for the phonons and the optical magnon. Non-degenerate phonon A-modes are in black, doubly-degenerate E-modes in light/dark color pairs, and the optical magnon in dashed dark blue around 28.47 meV. Non-degenerate phonon modes undergo no shift in frequency. Degenerate phonon modes show a slight splitting at the exception of the pair at 26.81 meV in violet/purple where the E_u^- mode shifts downward significantly in response to the coupling with the optical magnon. The optical magnon shifts upward by a similar amplitude. (b) Phonon angular momentum for each mode in the coupled regime. The non-degenerate phonon modes do not acquire any PAM. Degenerate phonons acquire significant PAM through the splitting in frequency. For each pair, the PAM has almost the same amplitude but with opposite signs. For the violet/purple pair, the PAM has still opposite signs but no longer the same amplitudes. This is a consequence of the phonon-magnon hybridization between the E_u^- and the optical magnon, where the optical magnon gains phononic character and steals PAM at the same time.

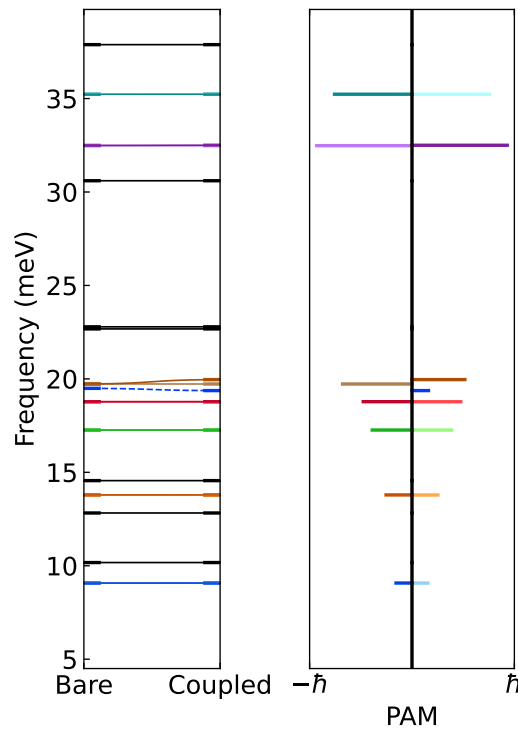


Figure 2: Frequency splitting and phonon angular momentum for CrBr₃ at the Γ point. (a) Bare and coupled frequencies for the phonons and the optical magnon. Non-degenerate phonon A-modes are in black, doubly-degenerate E-modes in light/dark color pairs, and the optical magnon in dashed dark blue around 19.49 meV. (b) Phonon angular momentum for each mode in the coupled regime. The non-degenerate phonon modes do not acquire any PAM. Degenerate phonons acquire significant PAM through the splitting in frequency.

Table 1: Bare frequencies, induced splitting/shift, and phonon angular momentum for the E phonon modes and the magnon modes for CrI_3 .

Irrep	E_0	ΔE	$L_z (\hbar)$
E_g	6.19	0.0079	.1401
			-.1412
	12.44	0.0001	.5277
			-.5272
	12.85	0.0045	.5332
			-.5327
	28.61	0.0047	.8539
			-.8539
	9.68	0.0022	.2550
			-.2548
E_u	13.31	0.0071	.7037
			-.7051
	26.81	0.1582	-.8824
			.9560
AM	0.80	0.0105	2.584×10^{-5}
OM	28.47	0.1566	-.0762

acter (see Table 4). The fact that only one phonon from the E_u pair hybridizes with the optical magnon is a cornerstone of chirality selective phonon-magnon coupling. Phonons and magnons with the same handedness will interact, whereas phonons and magnons of opposite handedness interact much more weakly [31, 33, 74, 75].

From Tables 1 and 2, we see that the angular momentum within one E pair of phonon modes almost adds up to zero and shows only a small imbalance. In contrast, the E_u pair that couples to the optical magnon show a much larger imbalance. This is explained by the fact that the E_u phonon that gains some magnonic character loses an equal amount of phononic character to the optical magnon, and with it some of its phonon angular momentum. As a consequence of gaining phononic character, the optical magnon also gains phonon angular momentum, which can be seen in Tables 1 and 2. During this transfer with the E_u mode, PAM is conserved within $\sim 10^{-3} \hbar$. When accounting for transfer between the optical magnon and all phonons, PAM is conserved within $\sim 10^{-4} \hbar$. The conservation of PAM is no longer an absolute rule due to a local metric modification by the mixed spin phonon Berry curvature.

4 Conclusion

Building upon the Lagrangian formulation of coupled spin-phonon dynamics, we derive a constrained Hamiltonian theory. This allows us to derive more symmetric and numerically more stable equations of motion. More significantly, this framework allows direct access to the total energy. By decomposing the total energy into phononic, magnonic and coupled contributions, we are able to quantify the degree of hybridization of phonon-magnon modes. We find phonon-magnon pairs with up to 8% and 25% hybridization for bulk CrI_3 and CrBr_3 respectively. We particularize our findings to pure non-time reversal symmetric phonons which include effects of the molecular Berry curvature but exclude the coupling with magnons. We show that the quantification of the hybridization through the decomposition of the total energy is a generalization to the decomposition of the norm of the displacement eigenvectors that is usually employed in the literature. We also derive an orthonormalization condition for the hybrid

Table 2: Bare frequencies, induced splitting/shift, and phonon angular momentum for the E phonon modes and the magnon modes for CrBr_3 .

Irrep	E_0 (meV)	ΔE (meV)	L_z ($\hbar$)
E_g	9.07	0.0004	.1555
			-.1557
	17.26	0.0002	.3884
			-.3883
	18.78	0.0008	.4761
E_u			-.4760
	35.23	0.0005	.7567
			-.7566
	13.79	0.0020	.2531
			-.2530
E_u	19.73	0.1212	-.6773
			.5138
	32.49	0.0038	-.9301
			.9303
AM	0.22	-0.0002	-0.6911E-06
OM	19.49	-0.1233	0.1632

Table 3: Bare frequencies, induced shifts, and decomposition into phonon, magnon and coupled character for selected E_u phonon modes and for the optical magnon in CrI_3 . The different characters are computed from Eq. (46) and are presented in %.

Irrep	E_0 (meV)	ΔE (meV)	$\eta^{(\text{ph})}$	$\eta^{(\text{mag})}$	$\eta^{(\text{cpld})}$
E_u^-	26.81	-0.1586	92.239	7.772	-0.011
E_u^+		0.0003	100.010	-0.010	0.000
OM	28.47	0.1566	7.828	92.161	0.011

phonon-magnon eigenvectors. This step had been overlooked up to now in the literature, but is crucial for the accurate computation of quantities relying on the mode eigenvectors. An example of this is the phonon angular momentum that we compute in both bulk CrI_3 and CrBr_3 . We find that, through hybridization with circularly polarized phonons, magnons are able to steal part of the phonon mechanical angular momentum. This stealing mechanism was not accounted for in current literature using naive orthonormalization schemes. In the future, we aim to extend the results obtained here to the whole Brillouin zone. This gives access to the full characterization of anti-crossing points between phonon and magnon bands. It allows the computation of band topology and topological transport as well [76].

An important characteristic of the current theory is that we do not employ the rigid spin approximation when incorporating adiabatic corrections. One could easily relax the atomic spin picture used here, as initially performed by Niu and Kleinman [42], to study materials with weakly localized 4d or 5d orbitals where the rigid atomic spin picture does not hold anymore [77]. The framework developed here can also be generalized and applied to other systems of composite hybrid particles such as magnon-plasmon [78, 79] and magnon-exciton hybrids [80].

We foresee that the theoretical tools developed in this paper will also prove useful to study the transport of phonon angular momentum in magnetic systems. These tools, and notably the stealing mechanism outlined in this paper, will enable the study of conversion of angular momentum between magnetic degrees of freedom and the underlying ionic lattice, for example

Table 4: Bare frequencies, induced shifts, and decomposition into phonon, magnon and coupled character for selected E_u phonon modes and for the optical magnon in CrBr_3 . The different characters are computed from Eq. (46) and are presented in %.

Irrep	E_0 (meV)	ΔE (meV)	$\eta^{(\text{ph})}$	$\eta^{(\text{mag})}$	$\eta^{(\text{cpld})}$
E_u^-	19.73	0.1200	74.984	25.007	0.009
E_u^+		-0.0012	100.003	-0.003	0.000
OM	19.49	-0.1233	25.047	74.962	-0.009

in ultrafast demagnetization processes [2]. The theory developed so far only deals with equilibrium states, and one would need to include scattering through higher order force constants, and coupling with external driving sources (light, electrical or spin currents, temperature gradient).

Funding information This work was supported by the Fonds National de la Recherche Scientifique (FRS-FNRS Belgium) through the Excellence of Science (EOS) program (Grant No. 40007563 CONNECT) and by the Fédération Wallonie Bruxelles through the ARC project DREAMS (G.A. 21/25-11). M.M. is an Aspirant PhD candidate of the Fonds de la Recherche Scientifique - FNRS, and received a Scientific Stay grant from the FNRS under number 2024/V 5/6/026 - JG/DeM - 2615 M.J.V. acknowledges funding by the Dutch Gravitation program “Materials for the Quantum Age” (QuMat, reg number 024.005.006), financed by the Dutch Ministry of Education, Culture and Science (OCW). M.R and M.S. acknowledge support from the Spanish MCIN/AEI/10.13039/501100011033 through grants PID2019-107338RB-C61, PID2022-139776NBC65, PRX22/00390, and PID2023-152710NB-I00 and a Severo Ochoa Excellence award to ICMAB, Grant CEX2023-001263-S, as well as by the Generalitat de Catalunya through Grant No. 2021 SGR 01519. Computational resources have been provided by PRACE (Optospin Project Id. 2020225411) on Discoverer at Sofiatech Bulgaria; by EuroHPC (Extreme Grant No. EHPCEXT-2023E02-050) on Marenosturm5 at Barcelona Supercomputing Center (BSC), Spain; on Lucia, the Tier-1 supercomputer of the Walloon Region, infrastructure funded by the Walloon Region under the grant agreement n°1910247; by the Dutch NWO for access to the LUMI supercomputer, owned by EuroHPC-JU and hosted by CSC (FI) through the ‘National Computer Facilities’ call (SURF reference EINF-16505).

A Comparison of bare frequencies with experimental values

In this appendix we compare the ab initio phonon frequencies obtained in the main text with those obtained experimentally in the literature. A first comparison of the computed bare phonon frequencies in CrI_3 against values obtained in Ref. [66] is shown in table 5. We use the bare phonon frequencies as to simplify the analysis since for the majority of the modes there is little change between the bare and the coupled frequencies. We also compare with other first-principles works. Overall, the obtained frequencies are in good agreement, at the exception of the A_g^2 mode which underestimate the measured frequency by about 20%. This can be explained by the fact that we employ an LDA exchange-correlation functional and that we did not use a Van der Waals correction. This is in contrast to Ref. [66, 67] where they use DFT-D Van der Waals corrections as well as GGA functionals.

In table 6, we compare the phonon frequencies obtained in the main text with experimental data from Ref. [68] and DFT calculations from Ref. [69]. The obtained frequencies matches well with the experiment although there is a noticeable deviation from the DFT calculations from Ref. [69] at higher frequencies.

Table 5: Comparison table for the bare phonon frequencies obtained for CrI_3 by solving Eq. (27) with experimental and numerical values from [66, 67]. All values are in cm^{-1} .

Irrep	w_0	Exp. [66]	DFT [66]	DFT [67]
E_g^1	49.9	54.1	59.7	53
A_u^1	54.2			64
A_g^1	69.8	73.3	89.6	79
E_u^1	78.1			83
A_g^2	85.8	108.3	98.8	88
E_g^2	100.3	102.3	99.8	98
E_g^3	103.6	106.2	112.2	102
E_u^2	107.4			107
A_g^3	124.9	128.1	131.1	125
A_u^2	125.2			123
A_g^4	206.0	-	195.2	195
E_u^3	216.2			206
E_g^4	230.8	236.6	234.4	225
A_u^3	249.1			240

B Additional data

In this appendix, we put the extended tables 7 and 8 regrouping the induced splitting, phonon angular momentum and hybridization characters for the phonon E modes and magnon modes for both bulk CrI_3 and CrBr_3 .

References

- [1] L. Zhang and Q. Niu, *Angular Momentum of Phonons and the Einstein–de Haas Effect*, Physical Review Letters **112**(8), 085503 (2014), doi:[10.1103/PhysRevLett.112.085503](https://doi.org/10.1103/PhysRevLett.112.085503).
- [2] S. R. Tauchert, M. Volkov, D. Ehberger, D. Kazenwadel, M. Evers, H. Lange, A. Donges, A. Book, W. Kreuzpaintner, U. Nowak and P. Baum, *Polarized phonons carry angular momentum in ultrafast demagnetization*, Nature **602**(7895), 73 (2022), doi:[10.1038/s41586-021-04306-4](https://doi.org/10.1038/s41586-021-04306-4).
- [3] C. Strohm, G. L. J. A. Rikken and P. Wyder, *Phenomenological Evidence for the Phonon Hall Effect*, Physical Review Letters **95**(15), 155901 (2005), doi:[10.1103/PhysRevLett.95.155901](https://doi.org/10.1103/PhysRevLett.95.155901).
- [4] A. V. Inyushkin and A. N. Taldenkov, *On the phonon Hall effect in a paramagnetic dielectric*, JETP Letters **86**(6), 379 (2007), doi:[10.1134/S0021364007180075](https://doi.org/10.1134/S0021364007180075).
- [5] G. Grissonnanche, S. Thériault, A. Gourgout, M.-E. Boulanger, E. Lefrançois, A. Ataei, F. Laliberté, M. Dion, J.-S. Zhou, S. Pyon, T. Takayama, H. Takagi *et al.*, *Chiral phonons in the pseudogap phase of cuprates*, Nature Physics **16**(11), 1108 (2020), doi:[10.1038/s41567-020-0965-y](https://doi.org/10.1038/s41567-020-0965-y).

Table 6: Comparison table for the bare phonon frequencies obtained for CrBr_3 by solving Eq. (27) with experimental and numerical values from [68, 69]. All values are in cm^{-1} .

Irrep	w_0	Exp. [68]	DFT [69]
E_g^1	73.2	74.7	71.3
A_u^1	82.0		83.8
A_g^1	103.4	107.0	105.5
E_u^1	111.2		111.0
A_g^2	117.4		116.2
E_g^2	139.2	143.4	137.3
E_g^3	151.5	153.3	141.8
E_u^2	159.1		149.9
A_u^2	182.9		174.9
A_g^3	183.8	184.7	177.3
A_g^4	246.8		222.7
E_u^3	262.0		240.2
E_g^4	284.2	285.0	266.3
A_u^3	305.6		290.1

- [6] G. Schaack, *Observation of circularly polarized phonon states in an external magnetic field*, Journal of Physics C: Solid State Physics **9**(11), L297 (1976), doi:[10.1088/0022-3719/9/11/009](https://doi.org/10.1088/0022-3719/9/11/009).
- [7] G. Schaack, *Magnetic field dependent splitting of doubly degenerate phonon states in anhydrous cerium-trichloride*, Zeitschrift für Physik B Condensed Matter **26**(1), 49 (1977), doi:[10.1007/BF01313371](https://doi.org/10.1007/BF01313371).
- [8] D. M. Juraschek, M. Fechner, A. V. Balatsky and N. A. Spaldin, *Dynamical multiferroicity*, Physical Review Materials **1**(1), 014401 (2017), doi:[10.1103/PhysRevMaterials.1.014401](https://doi.org/10.1103/PhysRevMaterials.1.014401).
- [9] S. Chaudhary, D. M. Juraschek, M. Rodriguez-Vega and G. A. Fiete, *Giant effective magnetic moments of chiral phonons from orbit-lattice coupling*, Physical Review B **110**(9), 094401 (2024), doi:[10.1103/PhysRevB.110.094401](https://doi.org/10.1103/PhysRevB.110.094401).
- [10] H. Chen, W. Wu, J. Zhu, Z. Yang, W. Gong, W. Gao, S. A. Yang and L. Zhang, *Chiral Phonon Diode Effect in Chiral Crystals*, Nano Letters **22**(4), 1688 (2022), doi:[10.1021/acs.nanolett.1c04705](https://doi.org/10.1021/acs.nanolett.1c04705).
- [11] S. Park and B.-J. Yang, *Phonon Angular Momentum Hall Effect*, Nano Letters **20**(10), 7694 (2020), doi:[10.1021/acs.nanolett.0c03220](https://doi.org/10.1021/acs.nanolett.0c03220).
- [12] D. A. B. Lopez, V. Brehm and D. M. Juraschek, *Atomistic theory of the phonon angular momentum Hall effect*, doi:[10.48550/arXiv.2604.01899](https://doi.org/10.48550/arXiv.2604.01899) (2026), [2604.01899](https://arxiv.org/abs/2604.01899).
- [13] C. P. Romao, R. Catena, N. A. Spaldin and M. Matas, *Chiral phonons as dark matter detectors*, Physical Review Research **5**(4), 043262 (2023), doi:[10.1103/PhysRevResearch.5.043262](https://doi.org/10.1103/PhysRevResearch.5.043262).

Table 7: Bare frequencies, induced splitting/shift, phonon angular momentum and phononic, magnonic and coupled characters for the E phonon modes and the magnon modes for CrI_3 .

Irrep	E_0	ΔE	$L_z (\hbar)$	$\eta^{(\text{ph})}$	$\eta^{(\text{mag})}$	$\eta^{(\text{cpld})}$
E_g	6.19	0.0087	.1399	100.043	-0.041	-0.002
E_g	6.19	0.0079	.1401	100.043	-0.043	-0.001
			-.1412	99.925	0.076	-0.001
	12.44	0.0001	.5277	100.000	0.000	0.000
			-.5272	100.000	-0.000	0.000
	12.85	0.0045	.5332	100.012	-0.012	0.000
			-.5327	99.986	0.014	0.000
	28.61	0.0047	.8539	100.000	-0.000	-0.000
			-.8539	100.000	0.000	-0.000
E_u	9.68	0.0022	.2550	99.987	0.013	0.000
			-.2548	100.003	-0.003	-0.000
	13.31	0.0071	.7037	99.923	0.079	-0.002
			-.7051	100.011	-0.011	0.001
	26.81	0.1582	-.8824	92.239	7.772	-0.011
			.9560	100.010	-0.010	0.000
AM	0.80	0.0105	2.584E-05	0.035	99.964	0.001
OM	28.47	0.1566	-.0762	7.828	92.161	0.011

- [14] T. Funato, M. Matsuo and T. Kato, *Chirality-Induced Phonon-Spin Conversion at an Interface*, Physical Review Letters **132**(23), 236201 (2024), doi:[10.1103/PhysRevLett.132.236201](https://doi.org/10.1103/PhysRevLett.132.236201).
- [15] H. Zhu, J. Yi, M.-Y. Li, J. Xiao, L. Zhang, C.-W. Yang, R. A. Kaindl, L.-J. Li, Y. Wang and X. Zhang, *Observation of chiral phonons*, Science **359**(6375), 579 (2018), doi:[10.1126/science.aar2711](https://doi.org/10.1126/science.aar2711).
- [16] Q. Cai, O. Hellman, B. Wei, Q. Sun, A. H. Said, T. Gog, B. Winn and C. Li, *Direct Observation of Chiral Phonons by Inelastic X-ray Scattering*, doi:[10.48550/arXiv.2108.06631](https://doi.org/10.48550/arXiv.2108.06631) (2021), [2108.06631](https://arxiv.org/abs/2108.06631).
- [17] H. Ueda, M. García-Fernández, S. Agrestini, C. P. Romao, J. van den Brink, N. A. Spaldin, K.-J. Zhou and U. Staub, *Chiral phonons in quartz probed by X-rays*, Nature **618**(7967), 946 (2023), doi:[10.1038/s41586-023-06016-5](https://doi.org/10.1038/s41586-023-06016-5).
- [18] H. Zhang, N. Peshcherenko, F. Yang, T. Z. Ward, P. Raghuvanshi, L. Lindsay, C. Felser, Y. Zhang, J.-Q. Yan and H. Miao, *Measurement of phonon angular momentum*, Nature Physics **21**(9), 1387 (2025), doi:[10.1038/s41567-025-02952-3](https://doi.org/10.1038/s41567-025-02952-3).
- [19] S. Coh, *Classification of materials with phonon angular momentum and microscopic origin of angular momentum*, Physical Review B **108**(13), 134307 (2023), doi:[10.1103/PhysRevB.108.134307](https://doi.org/10.1103/PhysRevB.108.134307).
- [20] M. Hamada and S. Murakami, *Phonon rotoelectric effect*, Physical Review B **101**(14), 144306 (2020), doi:[10.1103/PhysRevB.101.144306](https://doi.org/10.1103/PhysRevB.101.144306).
- [21] J. Sonntag, S. Reichardt, B. Beschoten and C. Stampfer, *Electrical Control over Phonon Polarization in Strained Graphene*, Nano Letters **21**(7), 2898 (2021), doi:[10.1021/acs.nanolett.0c05043](https://doi.org/10.1021/acs.nanolett.0c05043).

Table 8: Bare frequencies, induced splitting/shift, phonon angular momentum and phononic, magnonic and coupled characters for the E phonon modes and the magnon modes for CrBr_3 .

Irrep	E_0	ΔE	$L_z (\hbar)$	$\eta^{(\text{ph})}$	$\eta^{(\text{mag})}$	$\eta^{(\text{cpld})}$
E_g	9.07	0.0004	.1555	100.001	-0.000	-0.000
			-.1557	100.001	-0.000	-0.000
	17.26	0.0002	.3884	100.000	0.000	0.000
			-.3883	100.000	-0.000	0.000
	18.78	0.0008	.4761	100.000	-0.000	0.000
			-.4760	100.000	0.000	0.000
	35.23	0.0005	.7567	100.000	-0.000	0.000
			-.7566	100.000	0.000	0.000
E_u	13.79	0.0020	.2531	99.983	0.017	0.000
			-.2530	100.000	-0.000	-0.000
	19.73	0.1212	-.6773	100.003	-0.003	0.000
			.5138	74.984	25.007	0.009
	32.49	0.0038	-.9301	99.981	0.019	0.000
			.9303	100.001	-0.001	0.000
AM	0.22	-0.0002	-0.6911E-06	0.000	100.000	0.000
OM	19.49	-0.1233	.1632	25.048	74.962	-0.009

- [22] K. Moseni, R. B. Wilson and S. Coh, *Electric field control of phonon angular momentum in perovskite BaTiO_3* , Physical Review Materials **6**(10), 104410 (2022), doi:[10.1103/PhysRevMaterials.6.104410](https://doi.org/10.1103/PhysRevMaterials.6.104410).
- [23] R. M. Geilhufe, V. Juričić, S. Bonetti, J.-X. Zhu and A. V. Balatsky, *Dynamically induced magnetism in KTaO_3* , Physical Review Research **3**(2), L022011 (2021), doi:[10.1103/PhysRevResearch.3.L022011](https://doi.org/10.1103/PhysRevResearch.3.L022011).
- [24] M. Basini, M. Pancaldi, B. Wehinger, M. Udina, V. Unikandanunni, T. Tadano, M. C. Hoffmann, A. V. Balatsky and S. Bonetti, *Terahertz electric-field-driven dynamical multiferroicity in SrTiO_3* , Nature **628**(8008), 534 (2024), doi:[10.1038/s41586-024-07175-9](https://doi.org/10.1038/s41586-024-07175-9).
- [25] J. Bonini, S. Ren, D. Vanderbilt, M. Stengel, C. E. Dreyer and S. Coh, *Frequency Splitting of Chiral Phonons from Broken Time-Reversal Symmetry in CrI_3* , Physical Review Letters **130**(8), 086701 (2023), doi:[10.1103/PhysRevLett.130.086701](https://doi.org/10.1103/PhysRevLett.130.086701).
- [26] Y.-T. Wang, P.-G. Luan and S. Zhang, *Coriolis force induced topological order for classical mechanical vibrations*, New Journal of Physics **17**(7), 073031 (2015), doi:[10.1088/1367-2630/17/7/073031](https://doi.org/10.1088/1367-2630/17/7/073031).
- [27] L. Sheng, D. N. Sheng and C. S. Ting, *Theory of the Phonon Hall Effect in Paramagnetic Dielectrics*, Physical Review Letters **96**(15), 155901 (2006), doi:[10.1103/PhysRevLett.96.155901](https://doi.org/10.1103/PhysRevLett.96.155901).
- [28] Yu. Kagan and L. A. Maksimov, *Anomalous Hall Effect for the Phonon Heat Conductivity in Paramagnetic Dielectrics*, Physical Review Letters **100**(14), 145902 (2008), doi:[10.1103/PhysRevLett.100.145902](https://doi.org/10.1103/PhysRevLett.100.145902).
- [29] B. Flebus, K. Shen, T. Kikkawa, K.-i. Uchida, Z. Qiu, E. Saitoh, R. A. Duine and G. E. W. Bauer, *Magnon-polaron transport in magnetic insulators*, Physical Review B **95**(14), 144420 (2017), doi:[10.1103/PhysRevB.95.144420](https://doi.org/10.1103/PhysRevB.95.144420).

- [30] S. Streib, N. Vidal-Silva, K. Shen and G. E. W. Bauer, *Magnon-phonon interactions in magnetic insulators*, Physical Review B **99**(18), 184442 (2019), doi:[10.1103/PhysRevB.99.184442](https://doi.org/10.1103/PhysRevB.99.184442).
- [31] M. Weißenhofer, P. Rieger, M. S. Mrudul, L. Mikadze, U. Nowak and P. M. Oppeneer, *Chiral Phonons Arising from Chirality-Selective Magnon-Phonon Coupling*, Physical Review Letters **135**(21), 216701 (2025), doi:[10.1103/j7bs-2zbx](https://doi.org/10.1103/j7bs-2zbx).
- [32] D. Saparov, B. Xiong, Y. Ren and Q. Niu, *Lattice dynamics with molecular Berry curvature: Chiral optical phonons*, Physical Review B **105**(6), 064303 (2022), doi:[10.1103/PhysRevB.105.064303](https://doi.org/10.1103/PhysRevB.105.064303).
- [33] S. Ren, J. Bonini, M. Stengel, C. E. Dreyer and D. Vanderbilt, *Adiabatic Dynamics of Coupled Spins and Phonons in Magnetic Insulators*, Physical Review X **14**(1), 011041 (2024), doi:[10.1103/PhysRevX.14.011041](https://doi.org/10.1103/PhysRevX.14.011041).
- [34] C. A. Mead and D. G. Truhlar, *On the determination of Born–Oppenheimer nuclear motion wave functions including complications due to conical intersections and identical nuclei*, The Journal of Chemical Physics **70**(5), 2284 (1979), doi:[10.1063/1.437734](https://doi.org/10.1063/1.437734).
- [35] C. A. Mead, *The geometric phase in molecular systems*, Reviews of Modern Physics **64**(1), 51 (1992), doi:[10.1103/RevModPhys.64.51](https://doi.org/10.1103/RevModPhys.64.51).
- [36] P. Dirac, *Lectures on Quantum Mechanics*, Belfer Graduate School of Science, Monograph Series. Dover Publications, ISBN 978-0-486-41713-4 (2001).
- [37] J. L. Anderson and P. G. Bergmann, *Constraints in Covariant Field Theories*, Physical Review **83**(5), 1018 (1951), doi:[10.1103/PhysRev.83.1018](https://doi.org/10.1103/PhysRev.83.1018).
- [38] J. D. Brown, *Singular Lagrangians, Constrained Hamiltonian Systems and Gauge Invariance: An Example of the Dirac–Bergmann Algorithm*, Universe **8**(3), 171 (2022), doi:[10.3390/universe8030171](https://doi.org/10.3390/universe8030171).
- [39] M. Born, *Der Elektronen- und Kernbewegung in Molekeln und Kristallen*, Nachrichten der Akademie der Wissenschaften in Göttingen. Mathematisch-Physikalische Klasse **6**, 1 (1951).
- [40] J. Moody, A. D. Shapere and F. Wilczek, *Adiabatic effective Lagrangians*, In *Geometric Phases In Physics*, vol. 5 of *Advanced Series in Mathematical Physics*, p. 160. World Scientific (1989).
- [41] M. Born and K. Huang, *Dynamical Theory of Crystal Lattices*, Oxford Classic Texts in the Physical Sciences. Oxford University Press, Oxford, New York, ISBN 978-0-19-850369-9 (1998).
- [42] Q. Niu and L. Kleinman, *Spin-Wave Dynamics in Real Crystals*, Physical Review Letters **80**(10), 2205 (1998), doi:[10.1103/PhysRevLett.80.2205](https://doi.org/10.1103/PhysRevLett.80.2205).
- [43] Q. Niu, X. Wang, L. Kleinman, W.-M. Liu, D. M. C. Nicholson and G. M. Stocks, *Adiabatic Dynamics of Local Spin Moments in Itinerant Magnets*, Physical Review Letters **83**(1), 207 (1999), doi:[10.1103/PhysRevLett.83.207](https://doi.org/10.1103/PhysRevLett.83.207).
- [44] T. Holstein and H. Primakoff, *Field Dependence of the Intrinsic Domain Magnetization of a Ferromagnet*, Physical Review **58**(12), 1098 (1940), doi:[10.1103/PhysRev.58.1098](https://doi.org/10.1103/PhysRev.58.1098).

- [45] S. C. Guerreiro and S. M. Rezende, *Quantum Theory of the Magnon-Phonon Interaction in a Time-Dependent Magnetic Field*, Revista Brasileira de Física **1**(2), 207 (1971).
- [46] A. Auerbach, *Interacting Electrons and Quantum Magnetism*, Graduate Texts in Contemporary Physics. Springer, New York, NY, ISBN 978-1-4612-6928-1 978-1-4612-0869-3, doi:[10.1007/978-1-4612-0869-3](https://doi.org/10.1007/978-1-4612-0869-3) (1994).
- [47] F. Utermohlen, *Spin-Wave Theory Using the Holstein–Primakoff Transformation* (2020).
- [48] A. Holz, *Phonons in a strong static magnetic field*, Il Nuovo Cimento B (1971-1996) **9**(1), 83 (1972), doi:[10.1007/BF02735509](https://doi.org/10.1007/BF02735509).
- [49] R. P. Erickson and D. L. Mills, *Thermodynamics of thin ferromagnetic films in the presence of anisotropy and dipolar coupling*, Physical Review B **44**(21), 11825 (1991), doi:[10.1103/PhysRevB.44.11825](https://doi.org/10.1103/PhysRevB.44.11825).
- [50] L. Udvardi, L. Szunyogh, K. Palotás and P. Weinberger, *First-principles relativistic study of spin waves in thin magnetic films*, Physical Review B **68**(10), 104436 (2003), doi:[10.1103/PhysRevB.68.104436](https://doi.org/10.1103/PhysRevB.68.104436).
- [51] S. Toth and B. Lake, *Linear spin wave theory for single-Q incommensurate magnetic structures*, Journal of Physics: Condensed Matter **27**(16), 166002 (2015), doi:[10.1088/0953-8984/27/16/166002](https://doi.org/10.1088/0953-8984/27/16/166002).
- [52] P. Brüesch, *Phonons, Theory and Experiments: Theory and Experiments*, Springer-Verlag, ISBN 0-387-17223-8 (1982).
- [53] X. Gonze and C. Lee, *Dynamical matrices, Born effective charges, dielectric permittivity tensors, and interatomic force constants from density-functional perturbation theory*, Physical Review B **55**(16), 10355 (1997), doi:[10.1103/PhysRevB.55.10355](https://doi.org/10.1103/PhysRevB.55.10355).
- [54] T. Nomura, X.-X. Zhang, S. Zherlitsyn, J. Wosnitza, Y. Tokura, N. Nagaosa and S. Seki, *Phonon Magnetochiral Effect*, Physical Review Letters **122**(14), 145901 (2019), doi:[10.1103/PhysRevLett.122.145901](https://doi.org/10.1103/PhysRevLett.122.145901).
- [55] P. Delugas, O. Baseggio, I. Timrov, S. Baroni and T. Gorni, *Magnon-phonon interactions enhance the gap at the Dirac point in the spin-wave spectra of CrI_3 two-dimensional magnets*, Physical Review B **107**(21), 214452 (2023), doi:[10.1103/PhysRevB.107.214452](https://doi.org/10.1103/PhysRevB.107.214452).
- [56] Q. Wang, S. Liu, M.-Q. Long and Y.-P. Wang, *Regulation of magnon-phonon coupling by phonon angular momentum in two-dimensional systems*, Physical Review B **108**(17), 174426 (2023), doi:[10.1103/PhysRevB.108.174426](https://doi.org/10.1103/PhysRevB.108.174426).
- [57] A. L. Bassant, M. F. van Willigen and R. A. Duine, *Phonon amplification via magnetoelastic Klein scattering*, Journal of Applied Physics **137**(9), 093908 (2025), doi:[10.1063/5.0245641](https://doi.org/10.1063/5.0245641).
- [58] Ph. Ghosez, E. Cockayne, U. V. Waghmare and K. M. Rabe, *Lattice dynamics of BaTiO_3 , PbTiO_3 , and PbZrO_3 : A comparative first-principles study*, Physical Review B **60**(2), 836 (1999), doi:[10.1103/PhysRevB.60.836](https://doi.org/10.1103/PhysRevB.60.836).
- [59] T. Qin, J. Zhou and J. Shi, *Berry curvature and the phonon Hall effect*, Physical Review B **86**(10), 104305 (2012), doi:[10.1103/PhysRevB.86.104305](https://doi.org/10.1103/PhysRevB.86.104305).

- [60] M. A. McGuire, H. Dixit, V. R. Cooper and B. C. Sales, *Coupling of Crystal Structure and Magnetism in the Layered, Ferromagnetic Insulator CrI₃*, *Chemistry of Materials* **27**(2), 612 (2015), doi:[10.1021/cm504242t](https://doi.org/10.1021/cm504242t).
- [61] M. A. McGuire, *Crystal and Magnetic Structures in Layered, Transition Metal Dihalides and Trihalides*, *Crystals* **7**(5), 121 (2017), doi:[10.3390/cryst7050121](https://doi.org/10.3390/cryst7050121).
- [62] X. Gonze, J. M. Beuken, R. Caracas, F. Detraux, M. Fuchs, G. M. Rignanese, L. Sindic, M. Verstraete, G. Zerah, F. Jollet, M. Torrent, A. Roy *et al.*, *First-principles computation of material properties: The ABINIT software project*, *Computational Materials Science* **25**(3), 478 (2002), doi:[10.1016/S0927-0256\(02\)00325-7](https://doi.org/10.1016/S0927-0256(02)00325-7).
- [63] M. J. Verstraete, J. Abreu, G. E. Allemand, B. Amadon, G. Antonius, M. Azizi, L. Baguet, C. Barat, L. Bastogne, R. Béjaud, J.-M. Beuken, J. Bieder *et al.*, *Abinit 2025: New capabilities for the predictive modeling of solids and nanomaterials*, *The Journal of Chemical Physics* **163**(16), 164126 (2025), doi:[10.1063/5.0288278](https://doi.org/10.1063/5.0288278).
- [64] J. P. Perdew and Y. Wang, *Accurate and simple analytic representation of the electron-gas correlation energy*, *Physical Review B* **45**(23), 13244 (1992), doi:[10.1103/PhysRevB.45.13244](https://doi.org/10.1103/PhysRevB.45.13244).
- [65] M. Royo and M. Stengel, *Dynamical Response of Noncollinear Spin Systems at Constrained Magnetic Moments*, *Physical Review X* **16**(1), 011049 (2026), doi:[10.1103/c68v-8tkb](https://doi.org/10.1103/c68v-8tkb).
- [66] S. Djurdjić-Mijin, A. Šolajić, J. Pešić, M. Šćepanović, Y. Liu, A. Baum, C. Petrovic, N. Lazarević and Z. V. Popović, *Lattice dynamics and phase transition in CrI_3 single crystals*, *Physical Review B* **98**(10), 104307 (2018), doi:[10.1103/PhysRevB.98.104307](https://doi.org/10.1103/PhysRevB.98.104307).
- [67] D. T. Larson and E. Kaxiras, *Raman spectrum of CrI_3 : An *ab initio* study*, *Physical Review B* **98**(8), 085406 (2018), doi:[10.1103/PhysRevB.98.085406](https://doi.org/10.1103/PhysRevB.98.085406).
- [68] D. P. Kozlenko, O. N. Lis, S. E. Kichanov, E. V. Lukin, N. M. Belozeroва and B. N. Savenko, *Spin-induced negative thermal expansion and spin-phonon coupling in van der Waals material CrBr₃*, *npj Quantum Materials* **6**(1), 19 (2021), doi:[10.1038/s41535-021-00318-5](https://doi.org/10.1038/s41535-021-00318-5).
- [69] Ł. Kipczak, A. Karmakar, M. Grzeszczyk, R. Janiszewska, T. Woźniak, Z. Chen, J. Pawłowski, K. Watanabe, T. Taniguchi, A. Babiński, M. Koperski and M. R. Molas, *Resonant Raman scattering of few layers CrBr₃*, *Scientific Reports* **14**(1), 7484 (2024), doi:[10.1038/s41598-024-57622-w](https://doi.org/10.1038/s41598-024-57622-w).
- [70] L. Chen, J.-H. Chung, B. Gao, T. Chen, M. B. Stone, A. I. Kolesnikov, Q. Huang and P. Dai, *Topological Spin Excitations in Honeycomb Ferromagnet CrI_3* , *Physical Review X* **8**(4), 041028 (2018), doi:[10.1103/PhysRevX.8.041028](https://doi.org/10.1103/PhysRevX.8.041028).
- [71] S. E. Nikitin, B. Fåk, K. W. Krämer, T. Fennell, B. Normand, A. M. Läuchli and Ch. Rüegg, *Thermal Evolution of Dirac Magnons in the Honeycomb Ferromagnet CrBr_3* , *Physical Review Letters* **129**(12), 127201 (2022), doi:[10.1103/PhysRevLett.129.127201](https://doi.org/10.1103/PhysRevLett.129.127201).
- [72] L. Ke and M. I. Katsnelson, *Electron correlation effects on exchange interactions and spin excitations in 2D van der Waals materials*, *npj Computational Materials* **7**(1), 4 (2021), doi:[10.1038/s41524-020-00469-2](https://doi.org/10.1038/s41524-020-00469-2).

- [73] T. Gorni, O. Baseggio, P. Delugas, I. Timrov and S. Baroni, *First-principles study of the gap in the spin excitation spectrum of the CrI_3 honeycomb ferromagnet*, Physical Review B **107**(22), L220410 (2023), doi:[10.1103/PhysRevB.107.L220410](https://doi.org/10.1103/PhysRevB.107.L220410).
- [74] A. G. Gurevich and G. A. Melkov, *Magnetization Oscillations and Waves*, CRC Press, Boca Raton, ISBN 0-8493-9460-0 (1996).
- [75] J. Cui, E. V. Boström, M. Ozerov, F. Wu, Q. Jiang, J.-H. Chu, C. Li, F. Liu, X. Xu, A. Rubio and Q. Zhang, *Chirality selective magnon-phonon hybridization and magnon-induced chiral phonons in a layered zigzag antiferromagnet*, Nature Communications **14**(1), 3396 (2023), doi:[10.1038/s41467-023-39123-y](https://doi.org/10.1038/s41467-023-39123-y).
- [76] E. Thingstad, A. Kamra, A. Brataas and A. Sudbø, *Chiral Phonon Transport Induced by Topological Magnons*, Physical Review Letters **122**(10), 107201 (2019), doi:[10.1103/PhysRevLett.122.107201](https://doi.org/10.1103/PhysRevLett.122.107201).
- [77] Y. Lin, X. Liu and J. Feng, *Assessing the atomic moment picture of spin dynamics with *ab initio* magnon wavefunctions*, Physical Review B **111**(5), 054419 (2025), doi:[10.1103/PhysRevB.111.054419](https://doi.org/10.1103/PhysRevB.111.054419).
- [78] A. T. Costa, M. I. Vasilevskiy, J. Fernández-Rossier and N. M. R. Peres, *Strongly Coupled Magnon–Plasmon Polaritons in Graphene-Two-Dimensional Ferromagnet Heterostructures*, Nano Letters **23**(10), 4510 (2023), doi:[10.1021/acs.nanolett.3c00907](https://doi.org/10.1021/acs.nanolett.3c00907).
- [79] T. Hirose, P. M. Gunnink and A. Mook, *Topological magnon-plasmon hybrids*, Physical Review B **113**(18), L180404 (2026), doi:[10.1103/PhysRevB.113.L180404](https://doi.org/10.1103/PhysRevB.113.L180404).
- [80] P. C. Adak, F. Dirnberger, S. Acharya, A. Kamra, X. Xu and V. M. Menon, *Excitons in van der Waals magnetic materials*, Nature Materials pp. 1–15 (2026), doi:[10.1038/s41563-026-02636-0](https://doi.org/10.1038/s41563-026-02636-0).